
Designing a Dashboard for Transparency and Control of Conversational AI

Yida Chen^{1*}, Aoyu Wu¹, Trevor DePodesta¹, Catherine Yeh¹, Kenneth Li¹,
Nicholas Castillo Marin¹, Oam Patel¹, Jan Riecke¹, Shivam Raval¹, Olivia Seow¹,
Martin Wattenberg^{1,2†}, Fernanda Viégas^{1,2†}

¹Harvard University

²Google Research

Abstract

Conversational LLMs function as black box systems, leaving users guessing about why they see the output they do. This lack of transparency is potentially problematic, especially given concerns around bias and truthfulness. To address this issue, we present an end-to-end prototype—connecting interpretability techniques with user experience design—that seeks to make chatbots more transparent. We begin by showing evidence that a prominent open-source LLM has a “user model”: examining the internal state of the system, we can extract data related to a user’s age, gender, educational level, and socioeconomic status. Next, we describe the design of a dashboard that accompanies the chatbot interface, displaying this user model in real time. The dashboard can also be used to control the user model and the system’s behavior. Finally, we discuss a study in which users conversed with the instrumented system. Our results suggest that users appreciate seeing internal states, which helped them expose biased behavior and increased their sense of control. Participants also made valuable suggestions that point to future directions for both design and machine learning research. The project page and video demo of our TalkTuner system are available at bit.ly/talktuner-project-page.

1 Introduction

Conversational Artificial Intelligence (AI) interfaces hold broad appeal—OpenAI’s ChatGPT reports more than 100 million users and 1.8 billion monthly page visits [44, 42]—but also have essential limitations. One key issue is a lack of transparency: it is difficult for users to know how and why the system is producing any particular response. The obvious strategy of simply asking the system to articulate its reasoning turns out not to work, since Large Language Models (LLMs) are highly unreliable at describing how they arrived at their own output, often producing superficially convincing but spurious explanations [48].

Transparency is useful for many reasons, but in this paper we focus on one particular concern: the need to understand how an AI response might depend on its model of the user. LLM-based chatbots appear to tailor their answers to user characteristics. Sometimes this is obvious to users, such as when conversing in a language with gendered forms of the word “you” [49]. But it can also happen in subtler, more insidious ways, such as “sycophancy,” where the system tries to tell users what they are likely to want to hear, based on political and demographic attributes, or “sandbagging,” where it may give worse answers to users who give indications of being less educated [40].

[†]Co-advisors. Work done at Harvard.

*Correspondence to: Yida Chen <yidachen@g.harvard.edu>, Fernanda Viégas <fernanda@g.harvard.edu>, Martin Wattenberg <wattenberg@g.harvard.edu>

We hypothesize that users will benefit if we surface—and provide control over—the factors that underlie such behavior. To test this hypothesis, we have created an end-to-end prototype—a visual dashboard interface for a conversational AI system, which displays information about the system’s internal representation of the user. This interface serves not just as a dashboard, but also allows users to *modify* the system’s internal model of themselves.

Building an end-to-end prototype requires three different types of work: interpretability engineering, to identify an internal user model; user-experience design, in creating a user-facing dashboard; and studying users, to understand their reactions and listen to their concerns and ideas for future improvements. For the first step, we based on work on LLaMa2Chat-13B, an open-source large language model optimized for chat [46]. Within the model’s activations, we identified approximate internal representations of four important user characteristics (age, gender, education level, and socioeconomic status) via linear probes (in a manner similar to [56]). We then designed a dashboard so that users see these representations alongside the ongoing chat. Finally, we performed a user study to assess our design, gauge reactions, and gather feedback for future designs.

Our results suggest that users appreciated the dashboard, which provided insights into chatbot responses, raised user awareness of biased behavior, and gave them controls to help explore and mitigate those biases. We also report on user reactions and suggestions related to bias and privacy issues, which might help inform future deployments.

2 Background and related work

Chatbot interfaces have been studied for decades [51], and their lack of transparency has been a perennial issue. When users interact with black-box algorithms they often develop “folk theories” to explain what they observe [17], and modern LLMs are no exception [15]. This tendency can lead to an overly high degree of trust in these systems [41]—an effect initially seen with a chatbot in the 1960s, ELIZA [51], and continuing in recent years [28]. One particular concern is the presence of bias in responses, which can be difficult to detect and thus may be accepted at face value [54].

One tempting way to understand a chatbot is to talk to it— i.e., simply ask for a natural-language explanation of its output. Unfortunately, current LLMs appear to be highly unreliable narrators, describing their reasoning in ways that are convincing yet spurious [48, 13] or even avoiding the question altogether. A more heavyweight approach is taken by tools that analyze LLM behavior to help developers search for bias [26] or make more general comparisons [5, 27]. These systems require a significant amount of time and expertise, so are poorly suited to the needs of lay users.

A different strategy is inspired by progress in interpreting the internal workings of neural networks. In particular, some evidence suggests that LLMs may contain interpretable “world models” which play an important role in their output (see [36] for a review). Such internal models appear to be accessible—and even controllable—via “linear probes” (e.g., [3, 29, 11, 32, 24]). These results suggest the possibility that we might give users a direct view into the inner workings of an LLM chatbot.

The idea of surfacing such data to end users in the form of an easy-to-read dashboard was raised in [49]. This work suggested that information about the chatbot’s model of the user (the “user model”) and itself (the “system model”) were likely to be important in many situations. A related proposal [56] suggested using “representation engineering” for similar purposes, based on extensive experiments using a probing methodology called “linear artificial tomography.” Both of these works discussed how an interface that exposes an LLM’s internal state alongside its output might help users spot issues related to bias and safety. Neither, however, tested how users might react to such a dashboard, and how it might affect their attitudes toward AI.

3 Overall design methodology

Our methodology is to build and study a “design probe” [25, 18]. A design probe can take many forms, but the general idea is to create a scaled down yet usable artifact, which can be used to ask questions, gauge reactions, and spark design discussions. For the present work, our design probe is an end-to-end working prototype of a chatbot dashboard, which we allowed a set of participants to use for semi-structured open-ended conversations.

The rest of the paper has two parts. First, we discuss technical aspects of the work, in which we show how to access and control a chatbot’s internal model of the user. Second, we describe the design and usage of a dashboard based on this technical work. Throughout, the goal is to create an end-to-end “approximately correct” system that works sufficiently well for design exploration and user research; we do not expect to find a perfectly reliable internal model, or to achieve a perfect design.

Historically, even imprecise instruments had value to early users. For example, before becoming stable and precise, early car gas gauges fluctuated wildly with motion [22]. Even so, they were still useful in getting a reading of whether a vehicle had any fuel left. For pilots, the imprecise early instruments in cockpits [16] were an important step towards eventually conducting instrumentation-aided flights at night and in poor visibility [52]. Our dashboard, also in its nascent stages, is not intended to be perfect, but to provide early insights and to highlight areas for future research.

4 Probes for identifying an internal user model

Table 1: Summary of synthetic conversation dataset. (See footnote on gender subcategories)

Attributes	Subcategories	# Convos	Consistency	Topics	Correlation
Age	Child (< 13), Adolescent (13 - 17), Adult (18 - 64), Older Adult (> 64)	4000	88%	171	0.0%
Gender	Male, Female	2400	93%	101	0.5%
Education	Some Schooling, High School, College & Beyond	4500	–	158	0.7%
SocioEco	Lower, Middle, Upper	3000	95%	109	1.3%

The first step in our process is to investigate whether the LLM has any representation of the user [49]. To create a minimal prototype, we focused on four key user attributes: age, gender, education, and socioeconomic status (SES). We selected these attributes because they are culturally central, and influence critical real-world decisions such as college admissions, hiring, loan approvals, and insurance applications [1, 43, 10, 6, 7, 45].

Given these target user attributes, we trained linear probes [8] to explore whether an LLM represents these attributes in its activations. For this purpose, each attribute was divided into discrete subcategories, which were probed separately². (See the “subcategories” column in Table 1.)

The training process requires two ingredients. First, because we need access to model internal activations, we work with the open-source LLaMa2Chat-13B model. Second, we need a training dataset. Acquiring this data is nontrivial, as we now describe.

4.1 Creating the conversation dataset

Training probes to identify user representations would ideally use a human/chatbot conversation dataset with labeled user information. Unsurprisingly, given our target attributes, such data was not readily available [23, 55, 14]. However, recent work has used LLMs to generate synthetic conversations [12, 30, 33]. Specifically, Wang *et al.* [50] showed that GPT-3.5 can accurately role-play various personalities. LLaMa2Chat [46] was also fine-tuned via LLM role-play. Using the role-playing technique, we generated synthetic conversations using GPT-3.5 and LLaMa2Chat.³ We used a similar approach to generate conversations for all target attributes (see Appendix A).

Quality of generated data: One may question the quality of the synthetic conversation data: do role-played users represent their assigned attribute and cover a range of topics? Manual inspection of 13,900 multi-turn conversations (average 7.5 turns) would be time-consuming and prone to human bias. Recent work [4, 19] suggests that more powerful LLMs like GPT-4 [2] surpass crowd workers in annotating textual data. We therefore opted to use GPT-4 to annotate the generated data.

²Initially, the dataset included non-binary as a gender subcategory. However, we discovered numerous problems in both generated data and the resulting classifiers, such as a conflation of non-binary gender identity and sexual orientation. Consequently, the non-binary category was removed. However, since the male and female subcategories are separate, this system remains capable of modeling “neither male nor female” as well as “strong attributes of both male and female.”

³For example, to generate conversations held with a male user, we used the following prompt: “Generate a conversation between a human user and an AI assistant. This human user is a male. Make sure the conversation reflects this user’s gender. Be creative on the topics of conversation.”